DSA SEMESTER PROJECT

Near-Duplicate Question Finder for Q&A Platforms

Detecting near-duplicate questions using Rolling Hash and Dynamic Programming

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Abstract

Q&A platforms often contain many duplicate or nearly identical questions, which are difficult to identify manually. This project proposes a system that uses k-shingling to break each question into smaller overlapping word groups and applies double polynomial rolling hashing to generate fingerprints for these groups. Hash-based filtering is then used to quickly shortlist candidate question pairs that may be duplicates. Finally, exact common-substring analysis, implemented using dynamic programming, verifies the true similarity between candidates. Together, these techniques aim to improve the efficiency and accuracy of duplicate question detection on Q&A platforms.

K-Shingling | **Rolling Hash** | **Hash Filtering** | **DP Verification**

Introduction

Problem

- Q&A websites contain many repeated or similar questions
- Different users ask the same question using different words
- Duplicate questions create redundant information
- Comparing every question to every other is inefficient

Objective

- Detect near-duplicate questions accurately
- Reduce unnecessary pairwise comparisons
- Verify possible duplicates with confidence

Example

Q1: *"How can I reverse a string in Java?"*

Q2: *"What is the best way to reverse a String using Java?"*

Both questions are nearly identical in meaning, even though the wording is different.

Software & Hardware Requirements

SW

Software Requirements

- Java programming language
- JDK 17 or later
- IDE: VS Code / IntelliJ IDEA / Eclipse
- Java Collections Framework
- CSV / TXT dataset for testing
- Windows, Linux, or macOS

HW

Hardware Requirements

- Intel Core i3 or equivalent processor
- 4 GB RAM minimum
- 10 GB free storage space
- Standard keyboard and mouse
- Monitor / display
- Internet connection (optional)

Literature Survey

Technique	Purpose	Limitation
TF-IDF	Measures word importance	Does not directly capture substring similarity
Jaccard Similarity	Compares sets of words	Can require many comparisons
Rabin-Karp	Uses hashing for pattern matching	Hash collisions may occur
Rolling Hash	Generates efficient fingerprints	Requires collision handling
Dynamic Programming	Finds common substrings	Can require more computation

Research Gap

Existing methods can either be computationally expensive or may produce false matches. This project combines rolling-hash-based filtering with exact substring verification to achieve efficient and reliable near-duplicate detection.

Methodology

1. **Input Questions**
   Raw questions collected from the Q&A platform
2. **Preprocessing**
   Convert to lowercase, remove unnecessary punctuation
3. **K-Shingling**
   Divide each question into overlapping k-word groups
4. **Double Polynomial Rolling Hash**
   Generate two hash values (fingerprints) per shingle
5. **Candidate Filtering**
   Match hash fingerprints to shortlist similar questions
6. **Exact Verification (DP)**
   Longest Common Substring using Dynamic Programming
7. **Duplicate / Not Duplicate**
   Final decision based on similarity threshold

Algorithms Used

1. **K-Shingling** — Break questions into overlapping k-word sequences
2. **Rabin-Karp / Rolling Hash** — Generate numerical fingerprints for the shingles
3. **Double Hashing** — Use two hash values to reduce the chance of collisions
4. **Dynamic Programming** — Find the Longest Common Substring between candidate questions
5. **Final Decision** — If similarity is above a chosen threshold, classify as near duplicates

Main DSA Concepts: Substring Search + Rabin-Karp / Rolling Hash + Dynamic Programming

THANK YOU

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Questions & Discussion